

Problem Set-2

Problem.1 Suppose within your web browser you click on a link to obtain a web page. Suppose that the IP address for the associated URL is not cached in your local host, so that a DNS look up is necessary to obtain the IP address. Suppose that n DNS servers are visited before your host receives the IP address from DNS; the successive visits incur a RTT of $RTT_1, \dots, RTT_n$. Further suppose that web page associated with the link contains exactly one object, a small amount of HTML text. Let RTT_0 denote the RTT between the local host and the server containing the object. Assuming zero transmission time of the object, answer the following:

- (a.) How much time elapses from when the client clicks on the link until the client receives the object.
- (b.) Suppose the page contains three very small objects. How much time elapses with
 - (i) Nonpersistent HTTP with no parallel TCP connections,
 - (ii) Nonpersistent HTTP with parallel TCP connections
 - (iii) Persistent HTTP with pipelining.

Problem.2 Consider a Institutional network having capacity of 10Mbps connected to internet through 1.5 Mbps access link. Consider average object size = 900,000 bits, and the average request rate from institution's browsers to origin servers = 1.5 requests/sec and delay from institutional access router to any origin server and back to router = 2 sec. For the average access delay, use $\Delta / (1 - \Delta\beta)$, where Δ is the average time required to send an object over the access link and β is the arrival rate of objects to the access link. Calculate the following

- (a.) Traffic intensity on the LAN and traffic intensity on the access link
- (b.) Total delay from host of Institutional Network to origin server to get the response.
- (c.) Now suppose a cache is installed in the Institutional LAN with hit rate of 40% then calculate same for the question given in part (b)

Problem.3 Suppose you open a startup company "startupx" and want to set up your company network. Your network has the following servers:

- DNS server: "dns1.startupx.com" with IP as "172.119.12.40"
 - Web server: "startupx.com" with two IP as "172.119.12.55" and "172.119.12.56". The web server also has a name as "www.startupx.com".
 - Email server: "mail.startupx.com" with IP as "172.119.12.60" Your Company's email address is "username@startupx.com".
- a.) What resource records (RRs) do you need to provide to the upper-level ".com" Registrar?
 - b.) What RRs do you need to put in your company's DNS server?